

Cloud-Native Healthcare Data Management System

Phase 1 — Problem Statement Definition & Justification

Group 23

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Problem Context & Motivation

Why Healthcare Needs Cloud-Native Infrastructure

Healthcare Industry Challenges

- ▶ India's healthcare generates 2+ exabytes of data annually, growing at 48% CAGR
- ▶ EHR adoption grew from 10% (2018) to ~40% (2025), creating massive data volumes
- ▶ 73% of healthcare organizations still rely on on-premise/hybrid infrastructure
- ▶ Telemedicine surged 500% post-COVID; IoT medical devices projected to hit 50B by 2030
- ▶ Manual scaling leads to 30-40% higher operational costs
- ▶ Data silos across hospital departments prevent unified patient care

Real-World Workplace Scenario (SL)

- ▶ Scenario: A mid-size hospital chain (5-10 branches) with fragmented on-premise IT
- ▶ Peak OPD hours cause system slowdowns — servers at 90%+ CPU utilization
- ▶ Patient records stored in branch-level databases — no unified view for doctors
- ▶ Seasonal spikes (flu, pandemics) require emergency hardware procurement
- ▶ Compliance audits (DISHA, ABDM) reveal gaps in encryption & access logging
- ▶ IT team spends 60% of time on maintenance vs. innovation

Survey of Existing Solutions

Literature Review & Current Landscape Analysis

On-Premise / Legacy

- Full control over hardware
- High CapEx, poor scalability
- Manual patching & upgrades
- Cannot handle traffic spikes
- Data silos across branches

Hybrid Cloud (Current Norm)

- Partial cloud for non-critical loads
- Complex multi-vendor management
- Inconsistent security policies
- Manual scaling, limited automation
- ~60% of hospitals use this model

Cloud-Native (Proposed)

- Microservices & auto-scaling
- Unified data layer, real-time access
- Built-in security & compliance
- Pay-per-use OpEx model
- Full automation & observability

Justification — Why Cloud-Native?

Bridging the Gap Between Current Systems and Industry Needs

Gap Analysis — Current System Problems

- ✗ Scalability Gap: Hybrid systems cannot auto-scale during peak demand (OPD hours, emergencies)
- ✗ Data Fragmentation: Patient records scattered across branch DBs — no single source of truth
- ✗ Security Gap: Manual access control, inconsistent encryption fails compliance audits
- ✗ Cost Inefficiency: Over-provisioned hardware idle 70% of the time; no pay-per-use model
- ✗ Innovation Gap: IT teams burdened with maintenance, unable to adopt AI/ML analytics

Cloud-Native — How It Solves This

- ✓ Auto Scaling: EC2 ASG handles traffic spikes with zero manual intervention
- ✓ Unified Data: Centralized RDS/DynamoDB with multi-AZ replication across branches
- ✓ Security by Design: IAM, KMS encryption, CloudTrail audit logs — compliance-ready
- ✓ Cost Optimization: Pay-per-use + Reserved Instances → ~40-60% cost reduction
- ✓ Innovation Ready: Lambda + Athena enable predictive analytics on patient data

Problem Statement

"A mid-size hospital chain in India operates on fragmented, on-premise IT infrastructure with partial cloud adoption. The system struggles with scalability during peak hours, data silos across branches, inconsistent security practices, and high operational costs. There is no cloud-native architecture to provide auto-scaling, unified patient data management, real-time analytics, and compliance-ready security — resulting in degraded patient care and operational inefficiency."

Situation Learning (SL) Alignment

Workplace Relevance

Mirrors real hospital
chain IT challenges

Open-Ended Problem

Multiple cloud solutions
can be evaluated

Data Sensitivity

HIPAA/DISHA compliance,
PHI data handling

Feasibility & Practicality

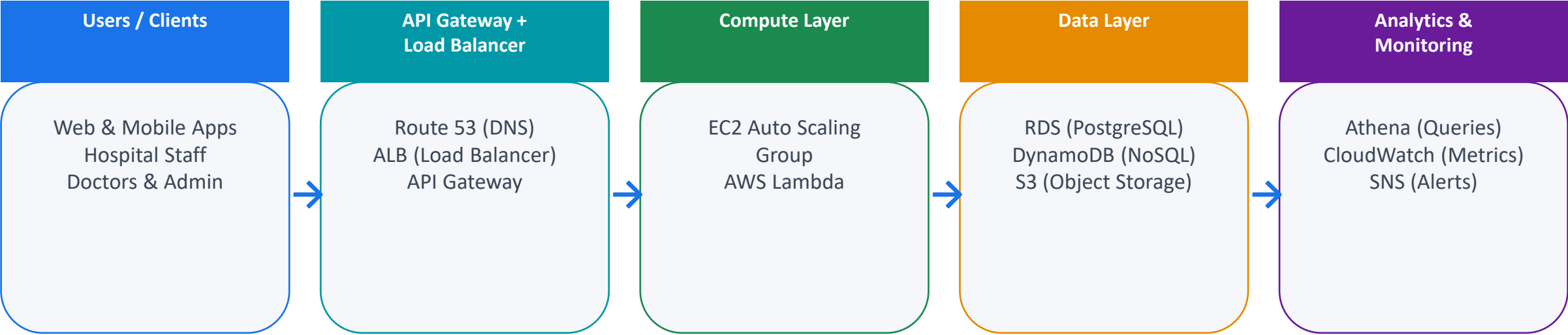
Managed AWS services,
realistic scope

Project Objectives

Clear, Measurable Goals Aligned with SL Guidelines

- O1** Design a cloud-native architecture using AWS (EC2, S3, RDS/DynamoDB, Lambda) that auto-scales to handle 3× peak traffic without manual intervention
- O2** Build a unified patient data management layer eliminating data silos across branches, achieving < 200ms query latency
- O3** Implement security & compliance framework — IAM role-based access, KMS encryption at rest/transit, CloudTrail audit logging aligned to HIPAA/DISHA
- O4** Deploy a serverless analytics pipeline (Lambda + S3 + Athena) for real-time patient flow monitoring and resource dashboards
- O5** Achieve 40-60% reduction in infrastructure costs by migrating to pay-per-use cloud model with Reserved Instances
- O6** Ensure 99.9% availability using Multi-AZ deployment, automated failover, and health checks via Elastic Load Balancer

Proposed Solution — High-Level Architecture



Security & Compliance Layer (Cross-Cutting)

IAM (Role-Based Access Control) · KMS (AES-256 Encryption at Rest & TLS in Transit) · VPC (Private Subnets & Network Isolation) · CloudTrail (API Audit Logs) · AWS Config (Compliance Rules) · WAF (Web Application Firewall) · MFA (Multi-Factor Authentication for Admin)

AWS Services & Technology Stack

Requirements for Problem Solving

Compute

- Amazon EC2 — Auto Scaling virtual servers
- AWS Lambda — Serverless event processing
- Elastic Load Balancer — Traffic distribution

Storage & Database

- Amazon S3 — Medical images, reports, backups
- Amazon RDS (PostgreSQL) — Patient records (OLTP)
- Amazon DynamoDB — Real-time session/IoT data

Networking & Security

- Amazon VPC — Network isolation & subnets
- AWS IAM — Role-based access policies
- AWS KMS — Encryption key management
- AWS CloudTrail — API audit logging

Analytics & Monitoring

- Amazon Athena — SQL queries on S3 data
- Amazon CloudWatch — Metrics & dashboards
- Amazon SNS — Notifications & alerts
- AWS Config — Compliance monitoring

Data Sensitivity & Compliance

SL Criterion: Secure Management of Sensitive Healthcare Data

Sensitive Data Types

- ⌘ Patient Health Information (PHI)
- ⌘ Electronic Health Records (EHR)
- ⌘ Medical imaging (X-rays, MRI scans)
- ⌘ Prescription & billing records
- ⌘ Doctor-patient communications
- ⌘ Insurance & financial data

Security Controls

- IAM: Least-privilege access policies
- KMS: AES-256 encryption at rest
- TLS 1.2+: Encryption in transit
- VPC: Private subnets for data layer
- CloudTrail: Complete API audit trail
- MFA: Multi-factor admin authentication

Compliance Standards

- ✓ HIPAA: US health data protection
- ✓ DISHA: India's health data law
- ✓ ABDM: Ayushman Bharat Digital Mission
- ✓ AWS Artifact: Compliance reports
- ✓ AWS Config: Automated compliance checks
- ✓ Regular security assessment cycles

Milestones & Development Journey

Phase-wise Breakdown with Timeline

Phase 1 17 Mar – 31 Mar	Problem Definition & Justification Define problem statement, survey existing solutions, identify AWS services, present Phase 1 ✓ Current
Phase 2A 03 Apr – 15 Apr	Architecture Design & Setup Detailed architecture design, AWS account setup, VPC & IAM configuration, security baseline Upcoming
Phase 2B 16 Apr – 28 Apr	Core Implementation Deploy EC2, RDS, S3, Lambda; build patient data APIs; integrate auto-scaling & monitoring Upcoming
Phase 2C 29 Apr – 03 May	Testing, Analytics & Submission Load testing, security audit, Athena analytics pipeline, record demo, submit deliverables Upcoming
Phase 3 04 May – 10 May	Final Presentation & Viva Refine presentation, prepare for Q&A, demonstrate working system, individual viva voce Upcoming

Roles & Responsibilities

Karan Rawat

System Design & Coordination

- Help design overall system flow (how components connect)
- Coordinate tasks among team members
- Assist in setting up basic cloud services
- Support integration of different modules

Vikas Kumar

Data & Backend Handling

- Work on understanding and structuring patient data
- Help design simple database (tables / storage idea)
- Assist in building basic APIs (if implemented)
- Support data storage and retrieval logic

Shared Roles

- learn AWS basics together
- Everyone will contribute to design discussions and decision making
- Each member will participate in testing, presentation, and documentation
- Regular collaboration to ensure complete understanding of the system

Kriti Tripathi

Security & Documentation

- Understand and define basic security requirements
- Document system design, workflows, and decisions clearly
- Ensure data privacy concepts are included
- Help in preparing reports and presentation content

Avanish Pratap Singh

Testing & Monitoring

- Work on testing system behavior (basic scenarios)
- Help track performance (like response time or system load conceptually)
- Assist in monitoring tools understanding
- Support final deployment and validation

All members will participate in all phases and maintain comprehensive understanding of the entire project.

Reflection & Expected Learnings

SL Criterion: Encouragement of Reflection

Workplace Relevance

This project mirrors real challenges in healthcare IT modernization. The experience of designing a cloud-native solution directly maps to skills needed in enterprise cloud migration projects — a growing demand in the industry.

Cloud Concepts to Learn

Auto Scaling, serverless computing (Lambda), managed databases (RDS/DynamoDB), security best practices (IAM, KMS), infrastructure monitoring (CloudWatch), cost optimization strategies, and compliance frameworks (HIPAA/DISHA).

Trade-offs & Decisions

Managed services vs. custom setup (RDS over self-managed DB for reduced ops). Multi-AZ for high availability vs. added cost. Serverless Lambda for event processing vs. always-on EC2. DynamoDB for speed vs. RDS for relational queries.

Thank You

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Questions & Discussion

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